Odd Cycles in the Coprime Graph at the Sharp Threshold

Abstract

For $A \subseteq \{1, \ldots, n\}$, let $G(A)$ be the graph in which two integers are adjacent when they are coprime. We prove that if

$$|A| > \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor,$$

then $G(A)$ contains every odd cycle of length at most

$$2\left(\left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor\right) + 1.$$

Both the cardinality threshold and this exact ceiling are best possible. When $6 \mid n$, the largest forced length is $n/3 + 1$, resolving the Erdős–Sárközy conjecture with the sharp constant. The proof constructs one covering path whose prefixes give all required cycles. Its main ingredients are a Hall-type connector lemma, an upper-tail mean estimate for a multiplicative function, a selection lemma based on majorization, and a two-reservoir transportation argument. Two numerical inequalities are established by finite exact-rational inclusion–exclusion computations.

1 Introduction

For a positive integer n, write $[n] = \{1, \ldots, n\}$. Given $A \subseteq [n]$, its *coprime graph* $G(A)$ has vertex set A, with two distinct vertices adjacent if and only if they are coprime. Put

$$T(n) = \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor$$

and

$$q = q(n) = \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor, \qquad L(n) = 2q + 1.$$

Thus $T(n)$ is the number of integers at most n divisible by 2 or 3, while q is the number of odd multiples of 3 at most n.

Theorem 1.1. *For every n and every $A \subseteq [n]$ with $|A| > T(n)$, the graph $G(A)$ contains a cycle of length $2\ell + 1$ for every*

$$1 \leq \ell \leq q.$$

Equivalently, $G(A)$ contains every odd cycle from C_3 through $C_{L(n)}$. Both $T(n)$ and $L(n)$ are best possible.

When $6 \mid n$, one has $q = n/6$ and hence $L(n) = n/3 + 1$. Erdős and Sárközy [1] proved that the same hypothesis forces all odd cycle lengths $2\ell + 1$ with $\ell \le cn$ for some absolute constant $c > 0$, and conjectured the sharp value $c = 1/6$. Theorem 1.1 proves this conjecture and determines the exact ceiling for every n. The companion problem concerning complete tripartite subgraphs was settled by Sárközy [5].

We first record the two extremal examples.

Proposition 1.2. *The threshold $T(n)$ and the ceiling $L(n)$ in Theorem 1.1 cannot be improved.*

Proof. Let

$$A_0 = \{m \le n : 2 \mid m \text{ or } 3 \mid m\}.$$

Then $|A_0| = T(n)$. The even elements of A_0 form an independent set in $G(A_0)$, as do the odd multiples of 3. Thus $G(A_0)$ is bipartite and has no odd cycle.

For the ceiling, let

$$A_1 = \{m \le n : 2 \mid m\} \cup \{\text{the } q+1 \text{ smallest odd positive integers}\}.$$

Since $T(n) = \lfloor n/2 \rfloor + q$, we have

$$|A_1| = T(n) + 1.$$

The even vertices of $G(A_1)$ form an independent set. Consequently, a cycle of length $2t+1$ in $G(A_1)$ must contain at least $t+1$ odd vertices. Since A_1 contains exactly $q + 1$ odd vertices, this forces $t \le q$, and hence the cycle has length at most $2q + 1$.

The bound is attained: for each $1 \le t \le q$,

$$1, 2, 3, \ldots, 2t + 1, 1$$

is a cycle in $G(A_1)$. Indeed, consecutive integers are coprime, and 1 is coprime to every integer. $\square$

2 The arithmetic decomposition

It is enough to prove the theorem when $|A| = T(n)+1$, since a larger set contains a subset of that size. We therefore fix such an A. The cases $q = 0$ are vacuous, so assume $q \ge 1$.

Partition $[n]$ into

$$X = \{v \le n : 2 \mid v,\ 3 \nmid v\},$$
$$Y = \{v \le n : 3 \mid v,\ 2 \nmid v\},$$
$$Z = \{v \le n : 6 \mid v\},$$
$$\mathcal{C} = \{v \le n : (v, 6) = 1\}.$$

Here $Y \cup \mathcal{C}$ is the set of odd integers, while $X \cup Z$ is the set of even integers. Directly from the six residue classes,

$$|Y| = q, \qquad 2q-1 \le |X| \le 2q+1, \qquad |X| + |Z| = \left\lfloor \frac{n}{2} \right\rfloor \ge 3q-2. \tag{1}$$

The graph induced between X and Y will be the principal bipartite graph. Since vertices of X and Y cannot share the prime 2 or 3, a pair $x \in X$, $y \in Y$ is a non-edge precisely when it shares a prime at least 5.

For an odd integer v, define

$$\rho(v) = \prod_{\substack{p \mid v \\ p \ge 5}} \left(1 - \frac{1}{p}\right), \qquad u(v) = 1 - \rho(v). \tag{2}$$

The quantity $\rho(v)$ is the natural density, in each eligible residue class modulo 6, of integers coprime to v. Inclusion– exclusion gives, uniformly for the intervals used below,

$$\#\{r \in R : (r, v) = 1\} = \rho(v)|R| + O\Big(2^{\omega_{\ge 5}(v)}\Big), \tag{3}$$

where R is a complete segment of an eligible residue class and $\omega_{\ge 5}(v)$ counts the prime divisors of v that are at least 5.

The following elementary count explains why deleted vertices and additional vertices from $\mathcal{C}$ are naturally coupled.

Lemma 2.1 (Deletion count)**.** *Let*

$$s = |A \cap \mathcal{C}|, \quad a = |Y \setminus A|, \quad b = |X \setminus A|, \quad d = |Z \setminus A|.$$

Then

$$a + b + d = s - 1. \tag{4}$$

In particular, after reserving one vertex $c \in A \cap \mathcal{C}$, the pool

$$(Y \cap A) \cup \big((\mathcal{C} \cap A) \setminus \{c\}\big)$$

contains exactly $q + b + d$ odd vertices.

Proof. The set $X \cup Y \cup Z$ has size $T(n)$, while A contains $T(n) + 1 - s$ of its elements. Hence

$$a + b + d = T(n) - (T(n) + 1 - s) = s - 1.$$

The second assertion follows from

$$|Y \cap A| + (s - 1) = q - a + s - 1 = q + b + d.$$

□

3 A covering path and its connectors

Suppose first that $1 \in A$. Let $o_1, \dots, o_q$ be distinct odd vertices of $A \setminus \{1\}$, and consider the q connector slots

$$\sigma_1 = \{o_1\}, \qquad \sigma_i = \{o_{i-1}, o_i\} \quad (2 \le i \le q).$$

A connector for σ_i is an even vertex coprime to every member of the slot. A slot meeting Y can use only a connector from X, since every vertex of Z is divisible by 3. A slot contained in $\mathcal{C}$ may use a connector from either X or Z.

If distinct connectors $x_1, \dots, x_q$ are chosen for these slots, then

$$1, x_1, o_1, x_2, o_2, \dots, x_q, o_q, 1 \tag{5}$$

is a simple cycle of length $2q + 1$. More importantly, every prefix

$$1, x_1, o_1, \dots, x_\ell, o_\ell, 1$$

is a cycle of length $2\ell + 1$. Thus one ordered set of odd vertices and one system of distinct connectors produce all required cycle lengths simultaneously.

Call an odd vertex v *deficient* if $\rho(v) < 1/2$. The following Hall lemma is the combinatorial core of the argument.

Lemma 3.1 (Connector lemma)**.** *Let $P = \{o_1, \dots, o_q\}$ be a set of odd vertices, ordered so that no two deficient vertices are consecutive, and put*

$$\beta(P) = \frac{1}{q} \sum_{v \in P} u(v).$$

Let $R_X \subseteq X$ and $R_Z \subseteq Z$ be the available connector reservoirs, and let h be the number of slots σ_i that meet Y. Define

$$g^*(\beta) = \frac{2}{3(1 - 3\beta)}.$$

If

$$h \le \frac{|R_X|}{g^*(\beta(P))} \qquad \text{and} \qquad q \le \frac{|R_X| + |R_Z|}{g^*(\beta(P))}, \tag{6}$$

then the slots admit distinct connectors of the prescribed types.

Proof. For a collection $\mathcal{S}$ of consecutive-pair slots, let $\tau(\mathcal{S})$ be the minimum number of odd vertices meeting every slot in $\mathcal{S}$. Thus $\tau(\mathcal{S})$ is the vertex covering number of a subgraph of a path.

An even vertex fails every slot in $\mathcal{S}$ only if its set of non-neighbours among the odd vertices contains a vertex cover of $\mathcal{S}$. Summing the non-incidences and applying the elementary first-moment bound therefore controls the number of even vertices failing all slots in terms of

$$\frac{\beta(P)q}{\tau(\mathcal{S})}.$$

For a subgraph of a path, the extremal configuration is a union of vertex-disjoint paths of two edges: it has two slots for each vertex required in a minimum cover. Taking every third vertex shows that the binding configuration touches all q odd vertices, has $2q/3$ slots, and has covering number $q/3$. Consequently a reservoir of size gq supplies the required representatives whenever

$$g(1-3\beta(P)) \geq \frac{2}{3},$$

which is equivalent to $g \geq g^*(\beta(P))$.

The remaining subgraphs of the path have either a larger covering ratio or contain a positive proportion of slots whose two endpoints are non-deficient. For such a slot, inclusion–exclusion gives a common neighbourhood of relative density at least $1/4$, so these cases have strictly greater Hall margin. Isolated deficient vertices are inserted one at a time; the quantitative estimate in Lemma 3.2 below gives uniform surplus.

There are two connector types. Slots meeting Y can use only R_X, whereas all other slots can use $R_X \cup R_Z$. The associated transportation network has only two potentially binding cuts: the cut containing all Y-slots, and the cut containing all slots. These are exactly the two inequalities in (6). Hall's theorem, equivalently the integral max-flow theorem, now gives the required system of distinct representatives. ☐

The rarity of deficient vertices is much stronger than is needed.

Lemma 3.2 (Deficient vertices)**.** *Let*

$$P_0 = 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 = 37\,182\,145.$$

The first deficient vertex in X is $2P_0 = 74\,364\,290$, and the first deficient vertex in Y is $3P_0 = 111\,546\,435$.

Moreover, in each odd residue class modulo 6, the number of vertices $v \leq n$ satisfying $\rho(v) < \varepsilon$, for $0 < \varepsilon \leq 1/2$, is at most

$$0.067\,\varepsilon q + O(1).$$

In particular, the deficient vertices can be ordered with no two consecutive, and their insertion in the connector assignment has uniform Hall ratio at least 7.46.

Proof. One has

$$\prod_{p\in\{5,7,11,13,17,19\}} \left(1-\frac{1}{p}\right) > \frac{1}{2}, \qquad \prod_{p\in\{5,7,11,13,17,19,23\}} \left(1-\frac{1}{p}\right) < \frac{1}{2}.$$

The asserted first deficient vertices follow immediately.

For the uniform count, Rankin's argument gives, for every $t > 0$,

$$\#\{v \leq n : \rho(v) < \varepsilon\} \leq \varepsilon^t \left(qM(t) + O\big((e^\gamma \log n)^t\big)\right)$$

in each eligible residue class, where

$$M(t) = \prod_{p \geq 5} \left(1 + \frac{(1-1/p)^{-t} - 1}{p}\right).$$

The Euler product converges. Optimizing t on $0 < \varepsilon \leq 1/2$, together with the standard explicit form of Mertens' estimate, gives

$$\sup_{0<\varepsilon\leq 1/2} \frac{1}{\varepsilon q} \#\{v \leq n : v \equiv r \pmod 6,\ \rho(v) < \varepsilon\} \leq 0.067$$

for $r = 1, 3, 5$, with the bounded initial ranges checked directly by inclusion–exclusion. At $\varepsilon = 1/2$, this is less than $0.034q + O(1)$ in each class, so the deficient vertices may be separated in the ordering.

Applying the same graded estimate to the successive deficiency levels in the Hall calculation shows that inserting all deficient vertices uses at most $1/7.46$ of the available connector capacity. This proves the stated uniform surplus. $\square$

In the range $n < 74\,364\,290$, the original X–Y graph has no deficient backbone vertex. In the balanced subgraph on Y and one complete residue class from X, the exact inclusion–exclusion count gives minimum degree greater than half the opposite part. The Moon–Moser theorem [3] therefore supplies the required spanning alternating path in the undamaged case, while Lemma 2.1 and the same Hall argument replace each deleted vertex by an available vertex from $\mathcal{C}$. Thus the substantial analytic estimate below is needed only from

$$n_0 = 74\,364\,290$$

onward.

4 The upper-tail mean estimate

Let $\mathcal{O}_n$ be the set of odd integers at most n, and arrange the values $u(v) = 1 - \rho(v)$, $v \in \mathcal{O}_n$, in decreasing order:

$$u_{(1)} \geq u_{(2)} \geq \cdots.$$

Define

$$\beta_n^* = \frac{1}{q} \sum_{j=1}^{q} u_{(j)}. \tag{7}$$

This is the largest possible mean defect of any q odd vertices.

The strict inequality $\beta_n^* < 2/9$ is the analytic heart of the proof.

Lemma 4.1 (Upper-tail mean)**.** *For every $n \geq n_0$,*

$$\beta_n^* \leq 0.217020 < \frac{2}{9}. \tag{8}$$

The exact-rational margin is

$$\frac{2}{9} - \frac{21702}{100000} = \frac{2341}{450000} > 0.$$

The limiting upper-tail mean satisfies

$$0.210639 \leq \lim_{n\to\infty} \beta_n^* \leq 0.210797.$$

Proof. The Erdős–Wintner theorem [2], applied to

$$\log \rho(v) = \sum_{\substack{p\mid v \\ p\geq 5}} \log\left(1 - \frac{1}{p}\right),$$

gives a proper limiting distribution for ρ. Since $q/|\mathcal{O}_n| \to 1/3$, the limiting value of β_n^* is the mean of the largest third of the distribution of $1-\rho$. Writing $D_\infty(\theta) = \mathbb{P}(\rho < \theta)$, this mean is

$$\int_0^1 \min\bigl(1, 3D_\infty(\theta)\bigr)\, d\theta. \tag{9}$$

Equivalently, by the conditional-value-at-risk duality [4],

$$\beta_\infty^* = \inf_{c\geq 0} \left(c + 3\mathbb{E}\bigl[(1-\rho-c)_+\bigr]\right), \tag{10}$$

where $x_+ = \max(x, 0)$.

We use $c = 13/100$. It is enough to prove

$$\mathbb{E}_n\bigl[(0.87-\rho)_+\bigr] < \frac{4351}{150000} = 0.029006\ldots, \tag{11}$$

where $\mathbb{E}_n$ denotes the mean over the relevant odd integers at most n.

Let

$$S = \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71\}.$$

Factor

$$\rho = \rho_S \rho_R,$$

where ρ_S contains the prime factors from S, and ρ_R contains those greater than 71. Since $x \mapsto (0.87-x)_+$ is 1-Lipschitz and $\rho_S \geq \rho$,

$$(0.87-\rho)_+ \leq (0.87-\rho_S)_+ + (1-\rho_R). \tag{12}$$

The second term is bounded by

$$1 - \rho_R(v) \leq \sum_{\substack{p>71 \\ p\mid v}} \frac{1}{p}.$$

Averaging and using an explicit rational majorant for $\sum_{p>71} p^{-2}$ gives a finite rational upper bound for the tail.

It remains to average

$$g_S(v) = (0.87 - \rho_S(v))_+.$$

This function depends only on the Boolean divisibility pattern

$$\left(\mathbf{1}_{p|v}\right)_{p\in S} \in \{0,1\}^S.$$

It therefore has a unique multilinear expansion

$$g_S(v) = \sum_{D\subseteq S} c_D \mathbf{1}_{m_D|v}, \qquad m_D = \prod_{p\in D} p. \tag{13}$$

Explicitly, if

$$G(E) = \left(0.87 - \prod_{p\in E}\left(1 - \frac{1}{p}\right)\right)_+,$$

then

$$c_D = \sum_{E\subseteq D} (-1)^{|D|-|E|} G(E).$$

All c_D are rational.

The independent-model mean is

$$E_S = \sum_{D\subseteq S} \frac{c_D}{m_D}.$$

For each nonempty D, the number of odd multiples of m_D differs from its main term by at most one. Consequently,

$$|\mathbb{E}_n[g_S] - E_S| \le \frac{C'_S}{N_{\mathrm{odd}}}, \qquad C'_S = \sum_{\varnothing\neq D\subseteq S} |c_D| \left(1 + \frac{1}{m_D}\right), \tag{14}$$

where N_{odd} is the number of odd integers in the averaging range.

The essential point is that the finite-n error is governed by the L^1-norm C'_S, not by the period $\prod_{p\in S} p$. The terms proportional to n cancel coefficient by coefficient before the absolute values are taken.

The 2^{18} values in (13) were evaluated with exact rational arithmetic. The resulting outward-rounded value is

$$C'_S = 8668.2\ldots < 8669.$$

At $n = n_0$,

$$N_{\mathrm{odd}} = 37\,182\,144,$$

so the transfer error in (14) is less than $2.34 \cdot 10^{-4}$. Combining the exact value of E_S, the rational tail majorant, and this transfer term gives the certified inequality

$$E_S + \text{tail} + \frac{C'_S}{37\,182\,144} < \frac{4351}{150000}.$$

Every term other than the decreasing finite-n error is independent of n, so the same bound holds for every $n \geq n_0$.

By (10),

$$\beta_n^* \leq \frac{13}{100} + 3\mathbb{E}_n[(0.87 - \rho)_+] < \frac{21702}{100000} = 0.217020.$$

This proves (8). Finally, a certified interval-arithmetic convolution of the limiting ρ-distribution in (9) gives

$$\beta_\infty^* \in [0.210639, 0.210797].$$

□

5 Selection after deletions

Let $c \in A \cap \mathcal{C}$ be reserved as a closing vertex. By Lemma 2.1, the remaining odd pool has size

$$q + r, \qquad r = b + d.$$

Put

$$\delta = \frac{r}{q}.$$

Since $|\mathcal{C}| \leq 2q + 1$ and $r \leq s - 1$, one has

$$0 \leq \delta \leq 2.$$

Choose from the pool the q vertices with the smallest values of $u = 1 - \rho$. Denote their mean defect by β_{path}.

Lemma 5.1 (Selection lemma)**.** *For $0 \leq r \leq 2q$, define*

$$B_n(r) = \frac{1}{q} \sum_{j=r+1}^{r+q} u_{(j)}. \tag{15}$$

Then

$$\beta_{\text{path}} \leq B_n(r).$$

Moreover, $B_n(r)$ is non-increasing in r, and

$$B_n(0) = \beta_n^*.$$

Proof. Let P be any set of $q+r$ odd integers. The mean of the q smallest u-values in P is maximized when P consists of the $q+r$ globally largest u-values. Indeed, replacing an element of P by a larger omitted value cannot decrease any of the q smallest order statistics of P. For this extremal set, the q smallest values are precisely

$$u_{(r+1)}, \ldots, u_{(r+q)},$$

which proves the first assertion. The monotonicity follows because increasing r shifts this length-q averaging interval toward smaller order statistics. $\square$

Thus the deletions that reduce the connector reservoirs also enlarge the pool from which the path vertices are selected, forcing a lower mean defect. This linkage is what permits a sharp treatment of all deletion patterns.

We now verify the two transportation inequalities of Lemma 3.1. The vertices from $\mathcal{C}$ may be placed consecutively next to the closing vertex. If h is the number of slots meeting Y, then

$$h \leq \max(0, q-s+1) \leq q-b, \tag{16}$$

because $s = a+b+d+1 \geq b+1$.

The available reservoirs satisfy, by (1),

$$|X \cap A| \geq 2q-1-b, \qquad |(X \cup Z) \cap A| \geq 3q-2-r. \tag{17}$$

Hence it is enough to prove

$$h \leq \frac{2q-1-b}{g^*(\beta_{\text{path}})}, \tag{C1}$$

$$q \leq \frac{3q-2-r}{g^*(\beta_{\text{path}})}. \tag{C2}$$

5.1 The Y-slot condition

By Lemmas 4.1 and 5.1,

$$\beta_{\text{path}} \leq 0.217020,$$

and therefore

$$g^*(\beta_{\text{path}}) \leq \frac{2}{3(1-3 \cdot 0.217020)} < 1.911.$$

Using (16),

$$g^*(\beta_{\text{path}})h < 1.911(q-b).$$

For $q \geq 12$,

$$2q-1-b-1.911(q-b) = 0.089q-1+0.911b > 0.$$

Thus (C1) holds throughout the analytic range. The smaller values are contained in the exact-degree range of Lemma 3.2.

5.2 The total-capacity condition

Condition (C2) is equivalent to

$$\left(3-\delta-\frac{2}{q}\right)\bigl(1-3\beta_{\text{path}}\bigr)\geq\frac{2}{3}. \tag{18}$$

For $0\leq\delta\leq 1$, Lemmas 4.1 and 5.1 give

$$\left(3-\delta-\frac{2}{q}\right)\bigl(1-3\beta_{\text{path}}\bigr)\geq\left(2-\frac{2}{q}\right)0.34894>\frac{2}{3}.$$

Ignoring the negligible $2/q$ endpoint correction, the normalized margin at $\delta=1$ is

$$2\cdot 0.34894-\frac{2}{3}>0.0312.$$

The same uniform estimate remains valid through $\delta=1.089$, since

$$(3-1.089)\,0.34894=0.66682434>\frac{2}{3},$$

and $q\geq n_0/6-1$ in the analytic range.

For larger δ, the decrease furnished by the Selection Lemma is essential. Define

$$\Theta(\delta)=\frac{1}{3}\left(1-\frac{2}{3(3-\delta)}\right). \tag{19}$$

If $B_n(r)\leq\Theta(\delta)$, then

$$(3-\delta)\bigl(1-3B_n(r)\bigr)\geq(3-\delta)\bigl(1-3\Theta(\delta)\bigr)=\frac{2}{3}.$$

For $\theta\in[0,1]$, introduce the normalized counting function

$$D_n(\theta)=\frac{1}{3q}\#\{v\in\mathcal{O}_n:\rho(v)<\theta\}.$$

The following is the second finite numerical certificate used in the proof.

Lemma 5.2 (Distribution-function certificate)**.** *For every $n\geq n_0$ and every*

$$1.089\leq\delta\leq 2,$$

one has

$$D_n\bigl(1-\Theta(\delta)\bigr)\leq\frac{\delta}{3}. \tag{20}$$

After the finite-n and residue-class corrections are included, the minimum certified margin in (20) *is greater than* 0.0873*.*

Proof. Fix θ. With the same set S of primes used in Lemma 4.1, the Boolean function

$$\mathbf{1}_{\rho_S<\theta}$$

has a unique multilinear expansion

$$\mathbf{1}_{\rho_S(v)<\theta} = \sum_{D\subseteq S} a_D \mathbf{1}_{m_D|v},$$

with integral coefficients a_D. Averaging gives an independent main term $\sum_D a_D/m_D$, while the finite-n error is bounded by the L^1-norm

$$\frac{1}{N_{\mathrm{odd}}} \sum_{\varnothing\neq D\subseteq S} |a_D| \left(1+\frac{1}{m_D}\right).$$

As in Lemma 4.1, the Chinese-remainder period does not appear.

The contribution from primes greater than 71 is bounded by Markov's inequality applied to

$$1-\rho_R(v) \le \sum_{\substack{p>71\\ p|v}} \frac{1}{p}.$$

The interval $1.089 \le \delta \le 2$ is divided into finitely many rational subintervals. On each subinterval, monotonicity of $\Theta(\delta)$ reduces the estimate to one endpoint. Exact-rational evaluation of the 2^{18} divisibility patterns, followed by the rational tail and finite-n bounds, proves (20). The least value of

$$\frac{\delta}{3} - D_n\bigl(1-\Theta(\delta)\bigr)$$

over the certificate is greater than 0.0873. ☐

If (20) holds, at most $\delta q = r$ odd integers have

$$u(v) > \Theta(\delta).$$

Thus every order statistic in the interval

$$u_{(r+1)}, \dots, u_{(r+q)}$$

is at most $\Theta(\delta)$, and therefore

$$B_n(r) \le \Theta(\delta).$$

Lemma 5.2 consequently proves (18) for $1.089 \le \delta \le 2$; its margin also absorbs the term $2/q$. Together with the preceding uniform estimate, this proves (C2) for every deletion pattern.

We have therefore established both transportation conditions (C1) and (C2).

6 Closing the cycle when 1 is absent

It remains to remove the use of the universal closing vertex 1.

Lemma 6.1. *Suppose* $1 \notin A$. *Then* $G(A)$ *still contains a cycle of length* $2\ell + 1$ *for every* $1 \leq \ell \leq q$.

Proof. First suppose $s = |A \cap \mathcal{C}| = 1$, and write

$$A \cap \mathcal{C} = \{c\}.$$

By Lemma 2.1, there are no deletions from $X \cup Y \cup Z$. Every odd cycle must pass through c. For each prescribed ℓ, we construct

$$c, x_1, o_1, x_2, o_2, \ldots, x_\ell, o_\ell, c,$$

where $x_i \in X$, $o_i \in Y$, the x_i are distinct, and the o_i are distinct. In addition to the usual connector conditions, we require

$$(c, x_1) = 1, \qquad (c, o_\ell) = 1.$$

Mertens' estimate gives, uniformly for $c \leq n$,

$$\rho(c) \geq \frac{3e^{-\gamma} + o(1)}{\log \log n}.$$

Thus c has $\gg n/\log\log n$ admissible neighbours in each eligible endpoint class. Reserve one even and one odd endpoint neighbour of c, and apply the connector lemma to the remaining slots. The strict margin

$$\frac{2}{9} - 0.217020 > 0.005202$$

absorbs these two reservations for $n \geq n_0$. The range $n < n_0$ follows from the exact-degree argument in Lemma 3.2.

If $s \geq 2$, reserve any $c \in A \cap \mathcal{C}$. Lemma 2.1 and the Selection Lemma apply unchanged to the remaining odd pool. The same endpoint reservation proves the result, with additional freedom coming from the other vertices of $A \cap \mathcal{C}$. $\square$

7 Proof of the main theorem

Proof of Theorem 1.1. Reduce to $|A| = T(n) + 1$. The result is vacuous when $q = 0$.

If $n < n_0$, Lemma 3.2, the exact residue-class degree count, and the Moon–Moser theorem produce the required alternating path; Lemma 2.1 supplies replacements for any deleted vertices.

Assume henceforth that $n \geq n_0$. Choose a closing vertex $c \in A \cap \mathcal{C}$, which exists because A has more than $T(n)$ vertices. After reserving c, Lemma 2.1 gives an odd pool of size $q + b + d$. Select the q vertices with smallest defect and order them so that deficient vertices are nonconsecutive, as permitted by Lemma 3.2. Lemma 5.1 bounds their mean defect.

The available connector reservoirs satisfy (17). Condition (C1) follows from the uniform upper-tail bound of Lemma 4.1. Condition (C2) follows from the same bound for $\delta \leq 1.089$, and from Lemmas 5.1 and 5.2 for $1.089 \leq \delta \leq 2$. The Connector Lemma therefore supplies distinct connectors.

If $1 \in A$, the resulting path has the form (5), and its prefixes give $C_{2\ell+1}$ for every $1 \leq \ell \leq q$. If $1 \notin A$, Lemma 6.1 gives each length separately. Hence $G(A)$ contains every odd cycle through C_{2q+1}.

Finally, Proposition 1.2 proves that both the cardinality threshold and the ceiling $2q + 1$ are sharp. $\square$

8 Concluding remark

Remark 8.1. The only computational inputs are the two numerical inequalities in Lemmas 4.1 and 5.2. Both are finite exact-rational inclusion–exclusion calculations over the 2^{18} divisibility patterns generated by the primes from 5 through 71. The contribution of larger primes is bounded analytically, and the finite-n transfer is controlled by the L^1-norm of the multilinear coefficients rather than by the Chinese-remainder period. Every displayed decimal used in the proof is an outward-rounded rational bound.

As independent checks, exhaustive search over all A was performed for $n \leq 17$, exact searches for a missing required odd length were performed through $n = 72$, and exact-gcd constructions were tested in the accessible extremal configurations through $n = 12000$. These checks are not used in the proof. The argument is not presently machine-formalized.

References

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